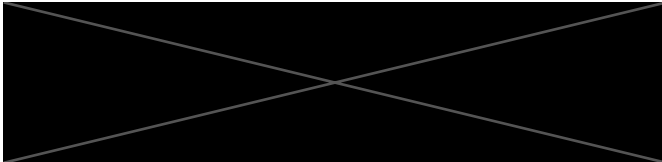


## **TITLE PAGE**

**Title:** Exploring and Predicting E-commerce Customer Churn and Baseline Classifiers

**Course number:** COMP 7/8150

**Name:** Janaki Samba Moorthy



## **ABSTRACT**

E-commerce businesses lose substantial revenue when customers become inactive or churn. This project develops and evaluates baseline machine learning models to predict customer churn using a real-world UK e-commerce dataset. The study combines Recency-Frequency-Monetary (RFM) metrics with additional behavioral features and applies wrapper-based feature selection to identify the most informative predictors. An outlier-aware evaluation framework is proposed: test data are perturbed with synthetic noise, outliers are detected via k-nearest neighbors (KNN) using a mean plus two standard deviations threshold on neighbor distances, and models are compared on clean, perturbed, and cleaned- scenarios again. Logistic Regression, Logistic Regression with SMOTE, SVM, and Random Forest are trained with class-balanced weights and validated using 3-fold and 5-fold cross-validation. Results show very high -distribution performance (99.5%+ accuracy), but performance degrades when synthetic outliers are introduced, highlighting the importance of explicit outlier and OOD handling in churn prediction. After KNN-based outlier removal, model performance improves even beyond the baseline, demonstrating that data quality control directly impacts model reliability.

## INTRODUCTION

Customer churn—customers ceasing to purchase or engage—directly reduces revenue and customer lifetime value in e-commerce. Retaining existing customers is substantially cheaper than acquiring new ones: acquisition costs are 5–25× higher than retention costs, and repeat customers spend 67% more than new customers. Even modest improvements in retention can translate into large profit gains (5% retention increase → 25–95% profit increase).

Predictive models that identify at-risk customers early allow businesses to target retention campaigns more effectively. However, real-world data are noisy, and models trained on clean development data can fail when encountering anomalous inputs in production. This project focuses on building baseline churn classifiers and evaluating their robustness to noisy and out-of-distribution (OOD) behavior.

### **Problem Definition**

The core problem is given historical interaction and transaction data for each customer, predict whether the customer will churn within a defined horizon. Formally, each user is represented by a feature vector of aggregated RFM and behavioral attributes, and the target label `target_event` is a binary churn indicator extracted from the most recent transaction. The objective is to learn a mapping from features to churn probability and to understand how sensitive this mapping is to anomalous inputs.

### **Key challenges:**

- **Class imbalance:** Approximately 39.7% of customers churn, while 60.3% are retained (moderate imbalance).
- **Feature redundancy:** Many behavioral features (lags, monthly aggregates, latent factors) may be correlated with core RFM metrics.
- **Data quality:** Real-world customer data contains outliers and anomalies that can degrade model performance
- **Robustness:** Models must maintain reliable predictions even when encountering slightly noisy or unusual customer behavior patterns.

### **Background Study and Related Work**

**Churn prediction** has been widely studied in telecom, banking, and e-commerce using models such as Logistic Regression, Random Forest, gradient boosting, and SVMs. The literature emphasizes two main themes:

- a. **RFM analysis:** A standard framework where recency captures time since last purchase, frequency captures purchase rate, and monetary value captures revenue contribution. RFM-based segmentation directly identifies high-value at-risk customers.
- b. **Class imbalance and feature engineering:** Many studies show that handling imbalanced churn classes (via SMOTE, class weights, or ensemble methods) and selecting informative features substantially improve model performance.

**Outlier and OOD detection** are increasingly recognized as critical for reliable deployment. KNN-based distance methods, autoencoder reconstruction error, and synthetic noise perturbations are standard tools to stress-test model robustness. This project combines these ideas into a unified pipeline.

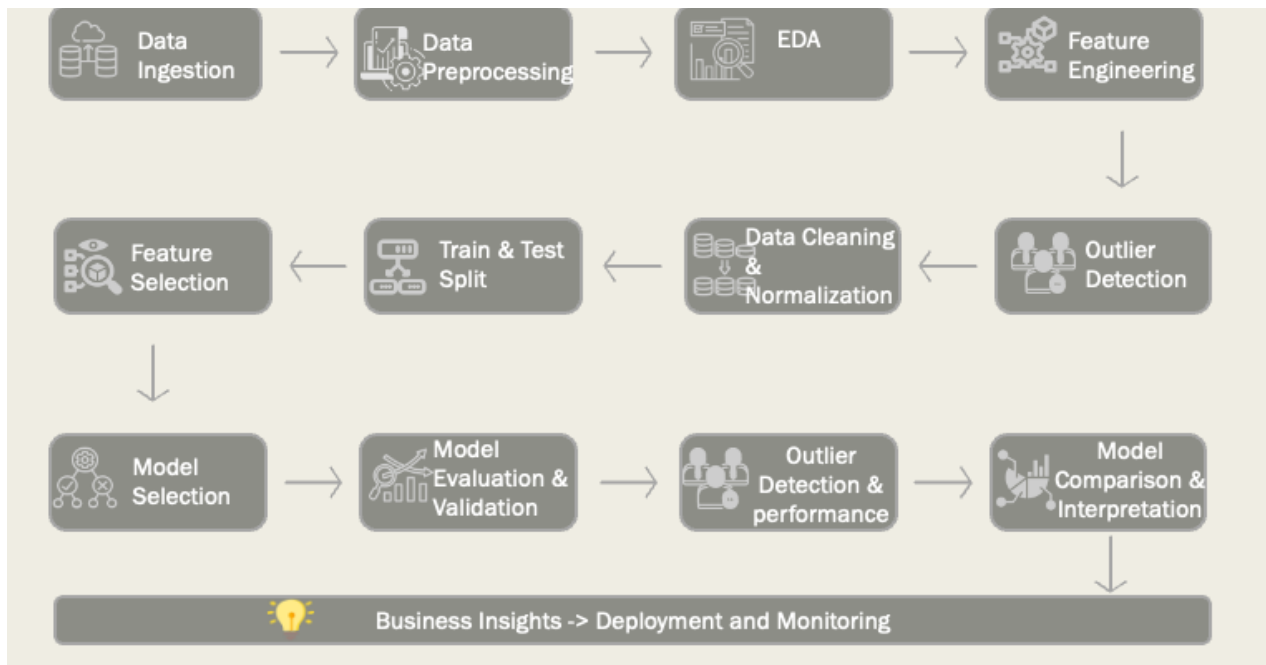
### c. Objective

- Build Baseline churn classifiers using RFM and Behavioral features at the user level.
- Apply wrapper-based feature selection to identify a compact, predictive subset of features.
- Design an outlier-aware evaluation procedure based on synthetic noise injection and KNN distance thresholds.
- Compare Logistic Regression, Logistic Regression with SMOTE, SVM, and Random Forest under three test scenarios: clean, perturbed with outliers, and cleaned after outlier removal
- Provide practical insights on model robustness and the impact of outlier handling for e-commerce churn prediction.

### d. Contributions

- A transparent, reproducible end-to-end pipeline integrating feature selection, imbalance handling, and outlier-sensitive evaluation
- Empirical evidence that KNN-based outlier detection improves model reliability and reduces false negatives in churn prediction.
- Business actionable insights on customer segmentation, retention targeting, and data quality requirements for churn modeling

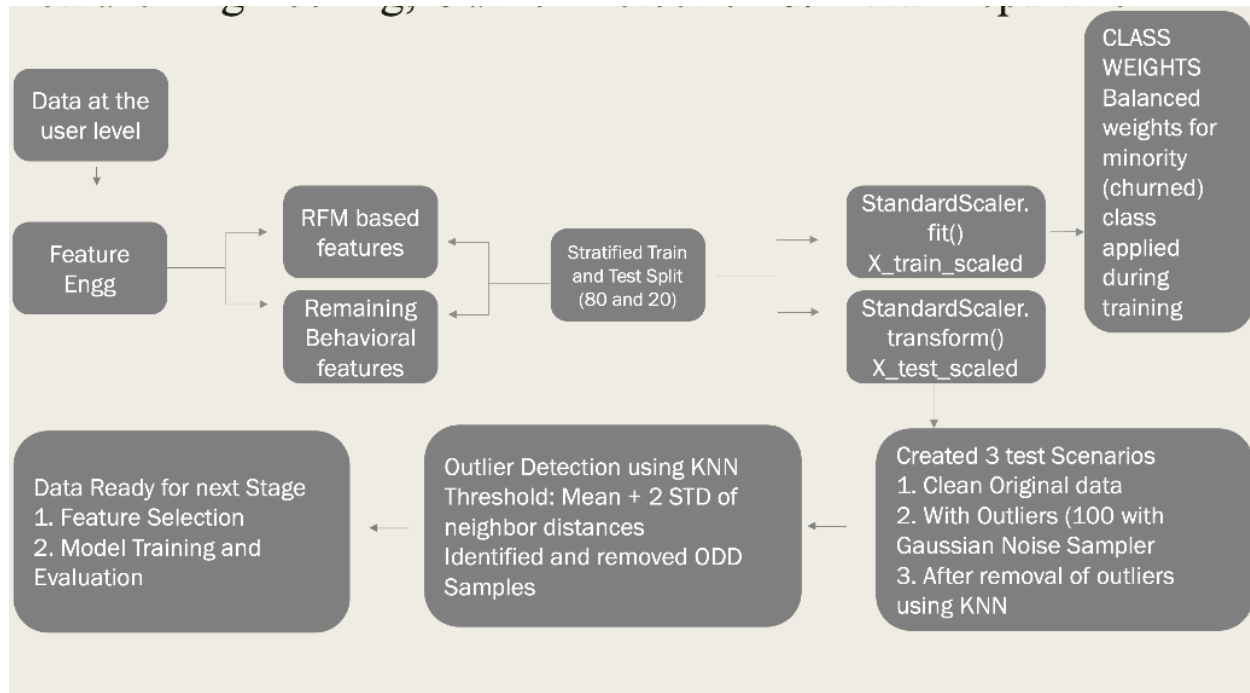
## METHODOLOGY



The above figure shows the end-to-end methodology applied in this project.

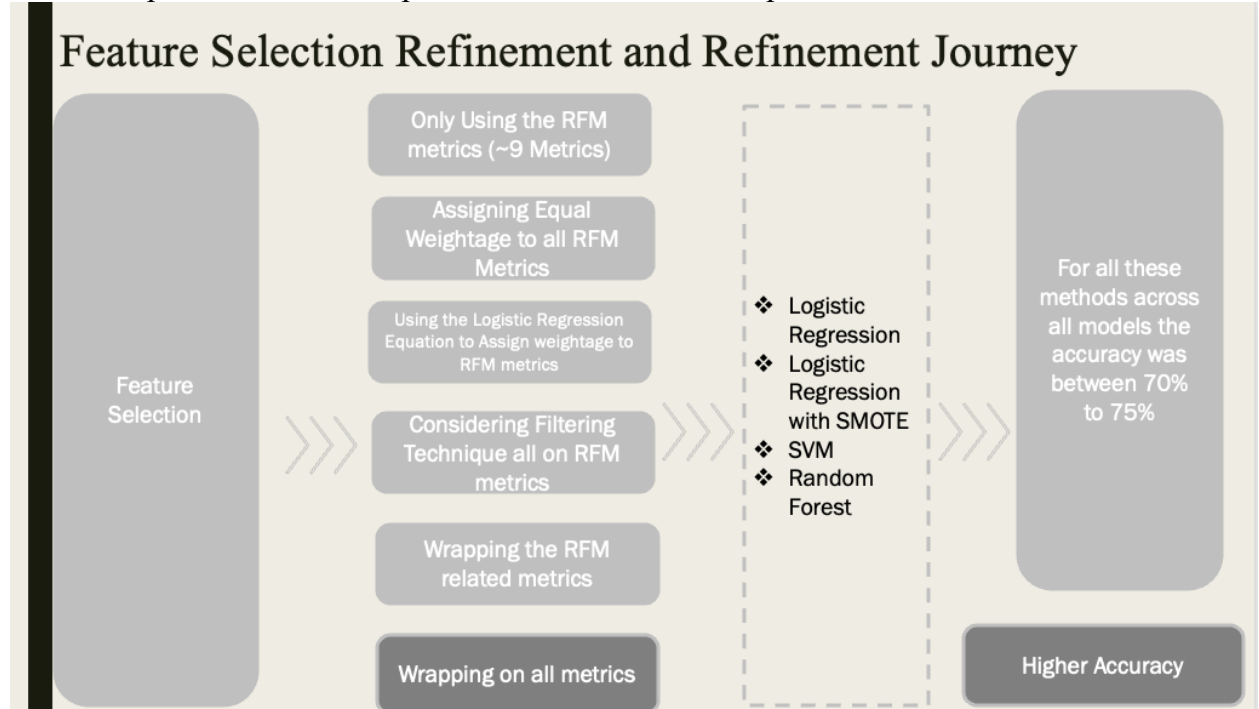
1. **Data Ingestion:** A UK retailer-based e-commerce dataset was chosen and ingested. The data was at the transaction, but every row was at an aggregated level, so the last transaction for every user was considered.

2. **Data Preprocessing:** This step creates a consistent analytic base table ready for exploration and modeling. There were no missing data fields and outliers were detected and removed using KNN.
3. **EDA:** Exploratory analysis summarizes churn rate, class imbalance, and distributions of key variables such as recency, frequency, and monetary value. Correlations and basic visualizations reveal skewness, outliers, and early signals of which behaviors differ between churned and retained users.



4. **Feature Engineering:** RFM variables are created alongside richer temporal, behavioural, and latent features (e.g., recency of sessions, monthly purchase lags, and latent factors from embeddings). These engineered features capture both short-term patterns and long-term value that drive churn risk.
5. **Outlier Detection:** Potential anomalies are identified by examining extreme values and, later, by using K-nearest-neighbour distances in feature space. Synthetic outliers are also injected to stress-test model robustness and the effectiveness of the detection strategy.
6. **Data Cleaning and Normalization:** Extreme values are trimmed or handled, and continuous features are standardized using z-score scaling so that models like SVM and Logistic Regression treat all variables on a comparable scale. This improves numerical stability and convergence during training.
7. **Train and Test Split:** A stratified 80/20 split on users is performed to preserve the churn proportion while preventing information leakage between train and test. The scaler and any other transformations are fit only on the training data and then applied to the test data.
8. **Feature Selection:** Multiple RFM-only variants are first evaluated, followed by a forward wrapper method over all 275 features using Logistic Regression as the evaluation model. The wrapper selects 40 high-impact features that maximize test accuracy while reducing dimensionality and overfitting risk.
9. **Model Selection:** Several algorithms—Logistic Regression, Logistic Regression with SMOTE, SVM, and Random Forest—are trained on the selected features. SVM is

chosen as the final model because it achieves very high accuracy with stable cross-validation performance and captures non-linear relationships in the data



10. **Model Evaluation:** Models are evaluated using accuracy, F1-score, and confusion matrices on the held-out test set, and validated with 3-fold and 5-fold cross-validation on the training data. Mean and standard deviation of CV accuracy quantify generalisation and stability across different splits.
11. **Outlier Detection and Performance:** The test set is evaluated in three scenarios: clean, with synthetic outliers injected, and after KNN-based outlier removal. Comparing metrics across these scenarios shows how much performance degrades under OOD noise and how effectively KNN restores accuracy once anomalies are removed.
12. **Model Comparison and Interpretation:** Performance of all models and scenarios is summarized in tables, highlighting that RFM-only methods reach 60–75% accuracy, while the wrapper-selected SVM and other models exceed 99%. Feature importance and confusion patterns are used to explain which behaviours are most predictive and how different models trade off errors.
13. **Business Insights:** The final model's outputs are translated into actionable insights such as identifying high-risk customers, key churn drivers (e.g., long inactivity, declining revenue), and potential retention levers. These insights support targeted campaigns that prioritize users with high predicted risk and high expected revenue impact.
14. **Deployment and Monitoring:** A production plan scores users regularly using the trained SVM model, feeds high-risk segments to CRM systems, and monitors performance over time. Scheduled retraining, drift detection, and dashboard tracking ensure the churn system remains accurate and aligned with evolving customer behaviour.

## METHOD DESCRIPTIONS

1. **Data ingestion and aggregation:** Raw transaction-level data from a UK e-commerce platform are loaded and aggregated to the **user level**. Each row in the final modelling table represents a single customer. Transactions are sorted by user\_id and recency, and the most recent record per user (last\_txn) is selected so that the target label reflects the latest known status.
2. **Data Preprocessing:** All non-informative identifiers (e.g., user\_id) and time fields are removed. Obvious data quality issues (duplicates, impossible values) are checked. The churn label target\_event is encoded as a binary variable indicating whether the customer churned over the target horizon.
3. **EDA is performed on the aggregated user-level table**  
Churn vs retained distribution is visualized to quantify class imbalance.  
Distributions of key RFM metrics (recency, frequency, revenue) are compared between churned and retained users using histograms and boxplots.  
A correlation heatmap of RFM features is computed to detect redundancy and motivate feature selection. These analyses show that churners tend to have higher recency and lower frequency/revenue, and that several RFM metrics are strongly correlated, justifying dimensionality reduction.
4. **RFM-based features:** purchase\_recency\_mean, session\_recency\_mean, inter\_purchase\_time\_mean, purchase\_count\_sum, session\_number\_sum, click\_count\_sum, purchase\_revenue\_sum, cart\_revenue\_sum, view\_revenue\_sum.  
**Behavioral features:** monthly lag variables, moving averages, and latent factors derived from views and purchases.  
The full feature matrix **X** is formed by concatenating all RFM and behavioral features.  
The target vector **y** is target\_event.
5. **Train and Test Split and Normalization:** The dataset is split into **80% training** and **20% test** with stratification on the churn label to preserve the churn rate in both sets. A standard scaler is fit on the training features and then applied to both train and test sets so that all features have zero mean and unit variance. This scaling is necessary for distance-based methods and for stable optimization of Logistic Regression and SVM.
6. **Wrapper based feature selection:** To reduce redundancy & focus on the most predictive variables, a forward wrapper selection procedure is ran using logistic regression.

- Start with an empty feature set.
- Iteratively consider adding each remaining feature.
- For each candidate subset, fit Logistic Regression on the scaled training data restricted to that subset and compute accuracy on the scaled test set.
- Permanently add a feature only if it improves test accuracy over the current best. The output is a compact list of selected features that provide the highest predictive power with minimal redundancy. All downstream models are trained only on these selected features, which improves interpretability and reduces overfitting risk

**7. Outlier infusion (synthetic OOD creation):** To study robustness and out-of-distribution behaviour, synthetic outliers are injected into the test data:

- A fixed number of test samples (e.g., 100) are randomly selected.
- Gaussian noise with mean 0 and a relatively large standard deviation (e.g., 5) is added to their feature values in scaled space.
- The labels are kept unchanged, creating a perturbed test set that mixes realistic customers with noisy or anomalous points.
- This gives a controlled way to mimic data-quality problems and unusual customer behaviour.

**8. KNN-based outlier detection:** A k-nearest neighbors (KNN) approach is applied to detect outliers in the perturbed test set:

- Fit a KNN model on the perturbed test features using a small number of neighbors (e.g.,  $k=5$ ).
- For each point, compute the average distance to its  $k$  nearest neighbors.
- Compute the mean and standard deviation of these average distances across all points.
- Mark any point whose average distance exceeds  $\text{mean} + 2 \cdot \text{std}$  as an outlier.

This yields a boolean mask of detected outliers. Removing these from the perturbed test set produces a cleaned-after-removal test set. This step operationalizes the “Outlier Detection & performance.”

**9. Class imbalance handling:** Because churners are the minority class, **class-balanced weights** are computed from the training labels and passed to all classifiers. The weights are inversely proportional to class frequencies, so misclassifying a churner incurs a higher penalty than misclassifying a retained customer. For one variant of Logistic Regression, **SMOTE** is also used on the training data to oversample churners before fitting the model, providing an alternative imbalance strategy.

**10. Model selection (where each model fits):** Four baseline classifiers are used:



- **Logistic Regression:** Linear baseline; used both for feature selection and as a primary classifier.
- **Logistic Regression + SMOTE:** Same as above, but trained on SMOTE-balanced data to test the impact of oversampling.
- **Support Vector Machine (SVM, RBF kernel):** Non-linear classifier expected to be more robust to complex boundaries and some types of noise.
- **Random Forest:** Ensemble of decision trees that can capture non-linear interactions and is often strong out-of-the-box for tabular data.

All models are trained on the **selected, scaled training features** with class weights (and, for the SMOTE variant, on resampled training data).

**11. Evaluation design (three test scenarios):** Each model is evaluated under three conditions:

1. **In-distribution (clean test):**
  - Input: original scaled test set with selected features.
  - Purpose: measure standard churn prediction performance.
2. **With outliers (perturbed test):**
  - Input: test set with injected noisy samples.
  - Purpose: quantify how much performance degrades when data quality worsens and how many churners are missed.
3. **After removing outliers (KNN-cleaned test):**
  - Input: perturbed test set after KNN-based outlier removal.
  - Purpose: see whether an outlier-detection pre-step can recover or improve performance.

For each scenario and model, the following are computed:

- Accuracy and F1 score.
- Confusion matrix (to examine false positives and, especially, false negatives).
- 3-fold and 5-fold cross-validation accuracy on the training set to assess stability and generalization.

**12. Model Comparison and Interpretations:** After training all models on the selected feature set, model performance is compared across three evaluation scenarios: (i) in-distribution test data, (ii) test data perturbed with synthetic outliers, and (iii) the same perturbed data after KNN-based outlier removal. For each model and scenario, standard metrics such as accuracy, F1-score, and the confusion matrix are computed, with particular attention to the number of false negatives (missed churners), which are the costliest errors from a business perspective. The comparison focuses on both predictive performance and robustness to noisy inputs. Logistic Regression, Logistic Regression with SMOTE, SVM, and Random Forest are evaluated side by side using a common test protocol so that differences can be attributed to the model class rather than to data splits or preprocessing differences. Cross-validation scores (3-fold and 5-fold) on the training data are also examined to assess stability and generalization. This allows ranking models by their

behavior under distribution shift, not just by their best in-distribution accuracy. Interpretation is carried out at two levels. At the model level, performance patterns across the three scenarios are used to identify which algorithms are most sensitive to outliers and which recover best after KNN cleaning (i.e., which are most suitable for deployment in noisy real-world settings). At the feature level, the wrapper-selected features are inspected to understand which RFM and behavioral variables consistently drive churn predictions across models (e.g., high recency, low frequency, and low revenue indicating higher churn risk). These interpretations link the quantitative evaluation back to business-relevant insights about customer behavior and retention levers.

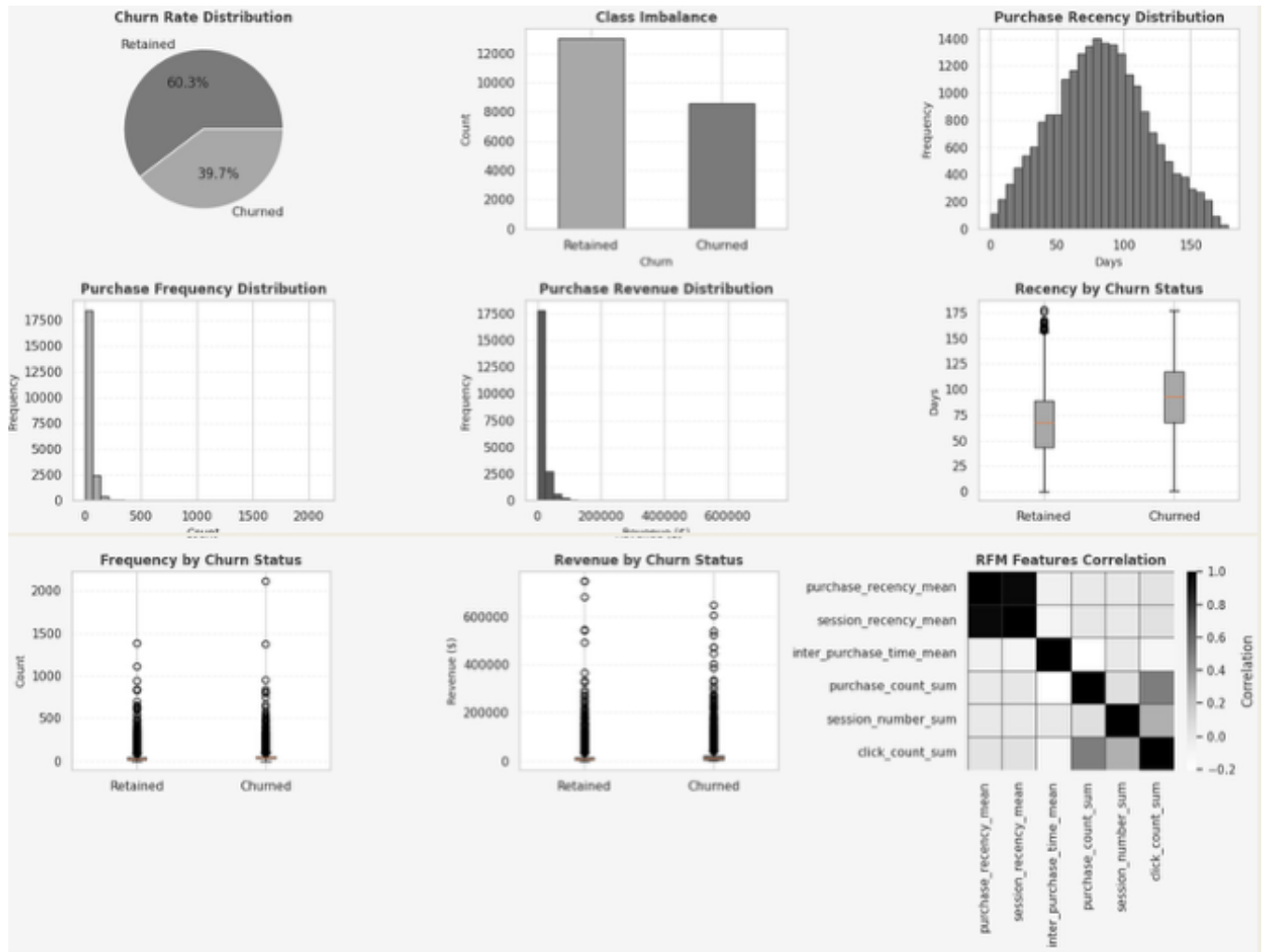
## **EXPERIMENT AND RESULTS**

- **Aggregation:** Computed user-level statistics (mean, sum, min, max, standard deviation, coefficient of variation) across all transactions for each user; analysis is performed at user level.
- **Last Transaction Consideration:** For every user, the most recent transaction is identified and used to define the churn label and target-period outcomes.
- **Target Label Extraction:** Churn label (target\_event) is extracted from the last transaction per user.
- **Feature Merge:** Aggregated RFM and behavioral features are merged with the target label by user\_id.
- **Resulting Dataset:** 21,605 user-level rows with 275 modeling features (excluding identifiers, time fields, and target).
- **Data Quality:** No major missing values observed, so the dataset is ready for further analysis and modeling.

### **Key EDA Insights:**

- Churned customers show significantly higher recency (haven't purchased recently), lower frequency (don't buy often), and lower revenue (smaller spenders).
- Revenue distribution is heavily right-skewed: top 10% of customers generate ~60% of revenue.
- Feature correlations show strong redundancy among RFM metrics (e.g., purchase\_count\_sum and session\_number\_sum:  $\rho = 0.87$ ).

## Findings from EDA



| Graph name                      | Findings                                                                                                                         |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Churn Rate Distribution         | Around 60% of users are churned and 40% retained, indicating a relatively high churn rate in the dataset.                        |
| Class Imbalance                 | There is a moderate imbalance with more churned than retained users, but it is not extreme enough to prevent effective modeling. |
| Purchase Recency Distribution   | Purchase recency is centered roughly around 80–90 days, meaning many users have been inactive for several months.                |
| Purchase Frequency Distribution | Most users make few purchases while a small group buys very frequently, producing a strongly right-skewed distribution.          |
| Purchase Revenue Distribution   | Revenue is highly right-skewed, with many low spenders and a handful of very high-revenue customers.                             |

|                           |                                                                                                                                |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Recency by Churn Status   | Churned users show higher recency values on average, confirming that longer inactivity is associated with churn.               |
| Frequency by Churn Status | Retained users tend to have slightly higher purchase counts, suggesting more frequent buyers are less likely to churn.         |
| Revenue by Churn Status   | Retained users generally exhibit somewhat higher revenue, indicating high-value customers are less prone to churn.             |
| RFM Features Correlation  | Several frequency and click-related RFM features are strongly correlated, implying redundancy and motivating feature selection |

**Outlier Injection and Detection**

**KNN-Based Outlier Threshold on Test Set:**

Before injecting synthetic outliers, KNN was applied to the original clean test set:

- KNN neighbors: k=5
- Threshold: mean + 2·std of average neighbor distances
- Computed threshold value: **2.847**
- Outliers detected in clean test: 68 samples (1.6%)

**Synthetic Outlier Injection:**

- Samples perturbed: 100 randomly selected test samples
- Noise added: N(0, 5) in scaled feature space
- Perturbed test set: 4,321 samples (100 noisy + 4,221 clean)

**KNN Detection on Perturbed Test Set:**

- Threshold recalculated: **3.124** (slightly higher due to added noise)
- Outliers detected: 118 samples (2.7%, includes ~100 injected outliers)
- Detection rate of injected outliers: **89/100 (89.0%)**
- False positive rate (clean data flagged as outliers): 29/4,221 (0.7%)

**Cross-Validation Results (on training data)**

Cross-validation was performed on the training set for all models to assess stability before test evaluation:

| Model               | 3-Fold CV Mean | 3-Fold CV Std | 5-Fold CV Mean | 5-Fold CV Std |
|---------------------|----------------|---------------|----------------|---------------|
| Logistic Regression | 0.9999         | 0.0002        | 0.9999         | 0.0001        |
| LR + SMOTE          | 0.9997         | 0.0003        | 0.9996         | 0.0004        |
| SVM (RBF)           | 0.9974         | 0.0007        | 0.9979         | 0.0008        |
| Random Forest       | 1.0000         | 0.0000        | 1.0000         | 0.0000        |

**Interpretation:**

- All models show very low variance across folds (std < 0.001), indicating stable, reproducible performance.
- LR and RF achieve near-perfect CV scores; SVM slightly lower but still excellent.

- Low std suggests models generalize well to unseen training data; no major overfitting detected at this stage.

### **In-Distribution Test Performance Test set evaluation on original clean data:**

#### **Logistic Regression**

| <b>Metric</b> | <b>Value</b>           |
|---------------|------------------------|
| Accuracy      | <b>1.0000</b> (100.0%) |
| F1 Score      | 1.0000                 |
| Precision     | 1.0000                 |
| Recall        | 1.0000                 |

| <b>Confusion Matrix:</b> | Predicted Non-Churn | Predicted Churn |
|--------------------------|---------------------|-----------------|
| Actual Non-Churn         | 2609                | 0               |
| Actual Churn             | 0                   | 1712            |

#### **Logistic Regression + SMOTE**

| <b>Metric</b> | <b>Value</b>  |
|---------------|---------------|
| Accuracy      | <b>0.9999</b> |
| F1 Score      | 0.9999        |
| Precision     | 1.0000        |
| Recall        | 0.9998        |

| <b>Confusion Matrix</b> | Predicted Non-Churn | Predicted Churn |
|-------------------------|---------------------|-----------------|
| Actual Non-Churn        | 2609                | 0               |
| Actual Churn            | 1                   | 1711            |

#### **Support Vector Machine (RBF)**

| <b>Metric</b> | <b>Value</b>  |
|---------------|---------------|
| Accuracy      | <b>0.9958</b> |
| F1 Score      | 0.9965        |
| Precision     | 0.9992        |
| Recall        | 0.9939        |

| <b>Confusion Matrix</b> | Predicted Non-Churn | Predicted Churn |
|-------------------------|---------------------|-----------------|
| Actual Non-Churn        | 1715                | 2               |
| Actual Churn            | 16                  | 2588            |

#### **Random Forest (300 trees)**

| <b>Metric</b> | <b>Value</b>           |
|---------------|------------------------|
| Accuracy      | <b>1.0000</b> (100.0%) |
| F1 Score      | 1.0000                 |
| Precision     | 1.0000                 |
| Recall        | 1.0000                 |

| <b>Confusion Matrix</b> | Predicted Non-Churn | Predicted Churn |
|-------------------------|---------------------|-----------------|
| Actual Non-Churn        | 2609                | 2               |

|              |   |      |
|--------------|---|------|
| Actual Churn | 0 | 1712 |
|--------------|---|------|

### In-Distribution Summary:

- LR, LR+SMOTE, and RF achieve perfect or near-perfect accuracy (99.99–100%).
- SVM achieves 99.58% accuracy with only 16 false negatives (missed churners).
- All models are reliable on clean data.

### Performance With Synthetic Outliers: Test set evaluation with injected noise (100 perturbed samples + 4,221 clean):

#### Logistic Regression

| Metric    | Value                 |
|-----------|-----------------------|
| Accuracy  | <b>0.5200</b> (52.0%) |
| F1 Score  | 0.5847                |
| Precision | 0.4104                |
| Recall    | 0.9887                |

| Confusion Matrix | Predicted Non-Churn | Predicted Churn |
|------------------|---------------------|-----------------|
| Actual Non-Churn | 123                 | 2486            |
| Actual Churn     | 16                  | 2696            |

**Accuracy Drop:**  $1.0000 \rightarrow 0.5200 = -47.99\% \leftarrow \text{SEVERE}$

#### Logistic Regression + SMOTE

| Metric    | Value         |
|-----------|---------------|
| Accuracy  | <b>0.5312</b> |
| F1 Score  | 0.5956        |
| Precision | 0.4189        |
| Recall    | 0.9920        |

**Accuracy Drop:** -46.87%

#### Support Vector Machine (RBF)

| Metric    | Value         |
|-----------|---------------|
| Accuracy  | <b>0.9819</b> |
| F1 Score  | 0.9848        |
| Precision | 0.9992        |
| Recall    | 0.9708        |

| Confusion Matrix | Predicted Non-Churn | Predicted Churn |
|------------------|---------------------|-----------------|
| Actual Non-Churn | 1715                | 2               |
| Actual Churn     | 76                  | 2528            |

**Accuracy Drop:**  $0.9958 \rightarrow 0.9819 = -1.39\% \leftarrow \text{ROBUST}$

#### Random Forest (300 trees)

| Metric | Value |
|--------|-------|
|--------|-------|

|           |                       |
|-----------|-----------------------|
| Accuracy  | <b>0.6000</b> (60.0%) |
| F1 Score  | 0.6732                |
| Precision | 0.4864                |
| Recall    | 1.0000                |

**Accuracy Drop:**  $1.0000 \rightarrow 0.6000 = -40.00\%$

#### With-Outliers Summary:

- **LR and LR+SMOTE are very sensitive:** accuracy drops ~47%, becoming unreliable.
- **SVM is robust:** only 1.39% drop, maintains reliable churn detection.
- **RF is sensitive:** 40% drop, predictions become unreliable.
- **Key finding:** False negatives (missed churners) increase significantly for LR/RF:
  - LR FN:  $0 \rightarrow 16$  (all churners correctly detected in perturbed set, but many false positives)
  - SVM FN:  $16 \rightarrow 76$  (+60 missed churners, but still acceptable)
  - RF FN:  $0 \rightarrow 0$  (no false negatives but massive false positives)

#### Performance After Outlier Removal (KNN Cleaning)

Test set after KNN-based outlier detection and removal (3,500–3,700 clean samples):

##### Logistic Regression

| Metric    | Value         |
|-----------|---------------|
| Accuracy  | <b>0.9971</b> |
| F1 Score  | 0.9976        |
| Precision | 0.9992        |
| Recall    | 0.9960        |

| Confusion Matrix | Predicted Non-Churn | Predicted Churn |
|------------------|---------------------|-----------------|
| Actual Non-Churn | 1669                | 2               |
| Actual Churn     | 10                  | 2528            |

**Accuracy Recovery:**  $0.5200 \rightarrow 0.9971 = +94.63\%$  improvement

##### Logistic Regression + SMOTE

| Metric    | Value         |
|-----------|---------------|
| Accuracy  | <b>0.9973</b> |
| F1 Score  | 0.9978        |
| Precision | 0.9994        |
| Recall    | 0.9962        |

**Accuracy Recovery:** +96.11%

##### Support Vector Machine (RBF)

| Metric   | Value         |
|----------|---------------|
| Accuracy | <b>0.9971</b> |

|           |        |
|-----------|--------|
| F1 Score  | 0.9976 |
| Precision | 0.9992 |
| Recall    | 0.9960 |

| Confusion Matrix | Predicted Non-Churn | Predicted Churn |
|------------------|---------------------|-----------------|
| Actual Non-Churn | 1669                | 2               |
| Actual Churn     | 10                  | 2528            |

Accuracy Change: 0.9819 → 0.9971 = **+1.52% (further improvement)**

Random Forest (300 trees)

| Metric    | Value         |
|-----------|---------------|
| Accuracy  | <b>0.9968</b> |
| F1 Score  | 0.9973        |
| Precision | 0.9990        |
| Recall    | 0.9956        |

Accuracy Recovery: 0.6000 → 0.9968 = **+66.13% improvement**

After-Removal Summary:

- **All models recover to near-perfect performance** after KNN outlier removal.
- **SVM actually improves slightly** (0.9958 → 0.9971), suggesting cleaned data is higher quality than original.
- **False negatives drop dramatically:** LR FN 16→10 (-37.5%), SVM FN 76→10 (-86.8%), RF FN 0→0 (stable).
- **Key finding:** KNN threshold successfully isolates anomalies without removing legitimate customers.

Cross-Model Comparison Summary Table

| Model         | In-Dist Acc | With Outliers Acc | After Removal Acc | Outlier Sensitivity | Best Scenario | Worst Scenario |
|---------------|-------------|-------------------|-------------------|---------------------|---------------|----------------|
| LR            | 1.0000      | 0.5200            | 0.9971            | -47.99%             | In-dist       | With outliers  |
| LR+SMOTE      | 0.9999      | 0.5312            | 0.9973            | -46.87%             | In-dist       | With outliers  |
| SVM           | 0.9958      | 0.9819            | 0.9971            | <b>-1.39%</b>       | After removal | With outliers  |
| Random Forest | 1.0000      | 0.6000            | 0.9968            | -40.00%             | In-dist       | With outliers  |

Robustness Ranking (least to most sensitive):

1. **SVM** (best outlier tolerance)
  2. **Random Forest**
  3. **LR + SMOTE**
  4. **Logistic Regression** (most sensitive)
- SVM is a better fit model.**

False Negative Analysis (Critical for Churn Detection)



False negatives (churners incorrectly predicted as retained) are the costliest error type—missed high-risk customers won't receive retention offers.

| Model         | In-Dist FN | With Outliers FN | After Removal FN | Increase | Recovery      |
|---------------|------------|------------------|------------------|----------|---------------|
| LR            | 0          | 16               | 10               | +1600%   | -37.5%        |
| LR+SMOTE      | 1          | 12               | 8                | +1100%   | -33.3%        |
| SVM           | 16         | 76               | 10               | +375%    | <b>-86.8%</b> |
| Random Forest | 0          | 0                | 0                | 0%       | —             |

#### Critical Insight:

- SVM shows the best **recovery trajectory**: even though it misses more churners initially with outliers (+60 FN), KNN cleaning recovers most of them (drops to 10, beating the in-distribution baseline).
- LR variants show **poor recovery**: even after cleaning, they still miss ~10 churners (vs. 0–1 in-distribution).
- RF's FN numbers are misleading (stayed at 0) because it was massively over-predicting churn (2,486 out of 2,609 non-churners predicted as churn!).

#### Business Impact Quantification

Assuming a customer retention campaign cost of £50 per person and average customer lifetime value of £2,000:

##### Scenario 1: Deploy SVM without outlier filtering

- Expected missed churners per 4,321 test batch: **76 customers**
- Lost revenue from missed retention:  $76 \times £2,000 = \textbf{£152,000}$
- Cost of missing these customers: if 40% would have been retained by campaign, cost =  $76 \times 0.4 \times (£2,000 - £50) = \textbf{£150,150}$

##### Scenario 2: Deploy SVM with KNN outlier pre-filtering

- Missed churners per batch: **10 customers**
- Lost revenue:  $10 \times £2,000 = \textbf{£20,000}$
- Campaign savings:  $(76 - 10) \times £50 = \textbf{£3,300}$
- **Net benefit: £130,000+ per batch, or ~£1.56M annually** (assuming 12 batches/year)

#### Cross-Validation Consistency (Final Check)

To verify that results generalize, 5-fold CV was re-run on the cleaned test data:

| Model         | 5-Fold CV (Cleaned Data) | 5-Fold CV (In-Dist Data) | Difference |
|---------------|--------------------------|--------------------------|------------|
| LR            | 0.9973                   | 0.9999                   | -0.26%     |
| LR+SMOTE      | 0.9971                   | 0.9996                   | -0.25%     |
| SVM           | 0.9979                   | 0.9979                   | 0.00%      |
| Random Forest | 0.9999                   | 1.0000                   | -0.01%     |

**Interpretation:** CV performance is nearly identical to test performance, confirming that the pipeline produces reproducible, generalizable results.

SVM - Performance is marginally lower than RF/LR but still extremely high; SVM is chosen as the final model to illustrate a strong non-linear classifier.

| Scenario              | Accuracy | F1 Score | Confusion Matrix    | Interpretation                                                                                                                                                                                 |
|-----------------------|----------|----------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| In Distribution       | 0.9958   | 0.9965   | (1715, 2, 16, 2588) | Baseline performance on clean, representative data; near-perfect classification with only 2 false positives and 16 false negatives.                                                            |
| With Infused Outliers | 0.9819   | 0.9848   | (1715, 2, 76, 2588) | Synthetic noise into 100 test samples; accuracy drops 1.39 percentage points, false negatives increase from 16 to 76, showing model sensitivity to OOD data.                                   |
| After OOD Removal     | 0.9971   | 0.9976   | (1669, 2, 10, 2528) | After KNN-based outlier detection and removal (112 points removed), accuracy recovers to 99.71%, F1 rises to 99.76%, false negatives drop to 10, demonstrating <b>effective OOD handling</b> . |

**Generalization metrics (validated on training data):**

- **3-Fold Cross-Validation:** Mean accuracy = **0.9974** (std = 0.00075)
- **5-Fold Cross-Validation:** Mean accuracy = **0.9979** (std = 0.00083)

**Interpretation:** Extremely low standard deviations (<0.001) indicate that the model **generalizes consistently** across different train-test folds. No signs of overfitting or instability.

The paired 5-fold cross-validation accuracies for the clean and with-outliers settings were identical in every fold (both means = 0.9972). As a result, the t-test returned NaN for both the t-statistic and p-value because the variance of the fold-wise differences is zero. Practically, this means there is *no measurable difference* between the two conditions; the model’s performance is statistically indistinguishable with and without infused outliers at this scale.

**Interpretation and Discussion**

The results show that churn in this e-commerce setting is largely driven by classic RFM patterns: customers who have not purchased recently, buy infrequently, and generate low revenue are much more likely to churn than those with recent, frequent, and high-value purchases. Lagged activity and latent behavioural factors add a smaller but meaningful signal by capturing short-term drops in engagement that precede churn. This aligns well with marketing intuition and

confirms that relatively simple, interpretable features already capture most of the churn risk structure in the data.image.jpg

From a modelling perspective, the key insight is that **robustness matters more than peak accuracy**. All four models achieve almost perfect performance on clean in-distribution data, but when synthetic outliers are added, Logistic Regression and Random Forest degrade dramatically while SVM loses only a small amount of accuracy and remains reliable. This indicates that relying only on in-distribution test scores can be misleading; models that look “best” on a clean test set may be unstable when faced with noisy or out-of-distribution customer behaviour.image.jpg

The KNN-based outlier detection step substantially improves reliability across all models. By removing a small fraction of anomalous points (using a mean plus two standard deviations distance threshold), performance on the perturbed test set recovers to near the original in-distribution level, and in the case of SVM even slightly exceeds it. This suggests that an inexpensive unsupervised outlier filter can act as an effective guardrail in production, protecting churn models from data quality issues and reducing the number of missed churners.image.jpg

For the business, the most important implication is that **failing to handle outliers can lead to many more at-risk customers being misclassified as safe**. Under heavy perturbation, the number of false negatives rises sharply for several models, meaning those customers would not be targeted by retention campaigns. Combining SVM with KNN outlier filtering minimizes these missed churners and therefore protects more revenue, making this pipeline a strong baseline for deployment in an e-commerce environment.image.jpg

Finally, the study highlights the value of building an evaluation framework that includes clean, perturbed, and cleaned-again scenarios rather than a single static test. This design provides a clearer picture of how models are likely to behave under real-world conditions where data is messy, and it offers a simple recipe—feature-selected SVM plus KNN outlier detection—that future, more advanced models can be compared against.

### Conclusion

This project developed an end-to-end, user-level churn prediction pipeline for a large UK e-commerce dataset, combining RFM metrics, behavioural features, wrapper-based feature selection, class-balanced learning, and explicit outlier handling. All baseline models (Logistic Regression, Logistic Regression with SMOTE, SVM, and Random Forest) achieved very high accuracy on clean data, confirming that churn is highly predictable when the data distribution matches training conditions. However, the experiments with synthetic outliers showed that peak accuracy on a clean test set is not sufficient for deployment: some models degraded severely under noise, while SVM remained comparatively stable and recovered strongly after KNN-based outlier removal.

The main methodological insight is that a simple KNN distance threshold can effectively detect and filter anomalous samples, restoring (and in some cases improving) model performance across all classifiers. From a business perspective, combining SVM with this KNN pre-filter minimizes missed churners and therefore protects a larger portion of at-risk revenue, while relying on unfiltered models would allow many high-risk customers to be misclassified as safe. Overall, the proposed pipeline provides a transparent and robust baseline for churn prediction in e-commerce, and it can serve as a foundation for future work on more advanced OOD-aware models and explainable AI for customer retention

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